

Integrated Circuit ECE326

Lec.07

Synthesis Digital Circuit

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Outlines



- ☐ DC transfer Ch/CS Curve
- ☐ Noise Margin Concept
- ☐ MOS Inverter Topologies
- ☐ DC Transfer Characteristics
- ☐ Beta Ratio Effects

Outlines



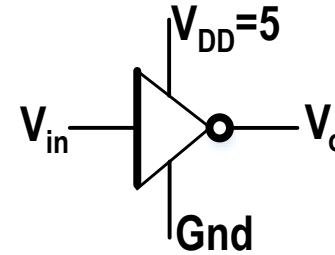
- ☐ **DC transfer Ch/CS Curve**
- ☐ **Noise Margin Concept**
- ☐ **MOS Inverter Topologies**
- ☐ **DC Transfer Characteristics**
- ☐ **Beta Ratio Effects**

DC transfer Ch/Cs curve



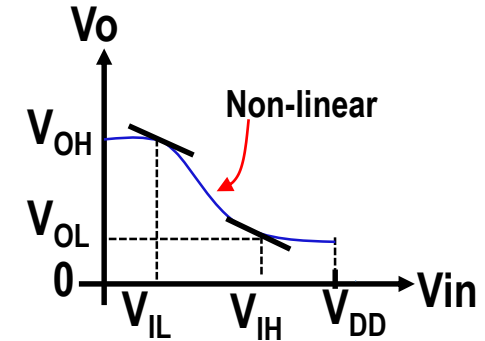
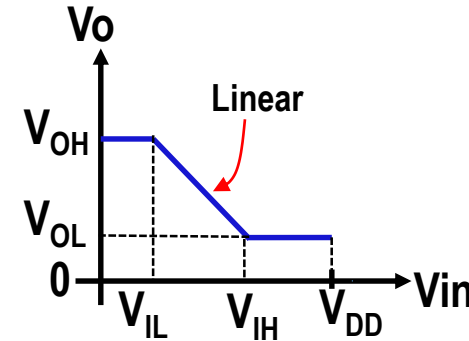
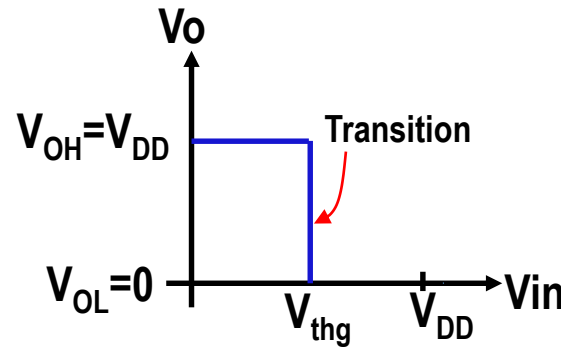
Definition

It is a relation between V_{in} and V_o



Types of Ch/Cs curve

- Ideal curve
- Simplified curve
- Practical curve



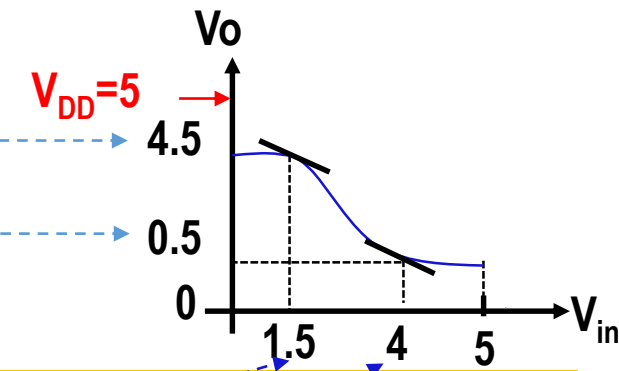
Notes

V_{OH} : Minimum output voltage can be considered as “1”

V_{OL} : Maximum output voltage can be considered as “0”

V_{IL} : Maximum input voltage can be considered as “0”

V_{IH} : Minimum input voltage can be considered as “1”



Ideally, $V_{OH} = V_{DD}$, $V_{OL} = \text{GND}$
 $V_{IH} = V_{IL} = V_{DD}/2$

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Noise Margin Concept (1/2)



Definition

It indicates how system can immune against the noise

Categories of Noise Margin

$NMH = V_{OH} - V_{IH}$

$NML = V_{IL} - V_{OL}$

Remember

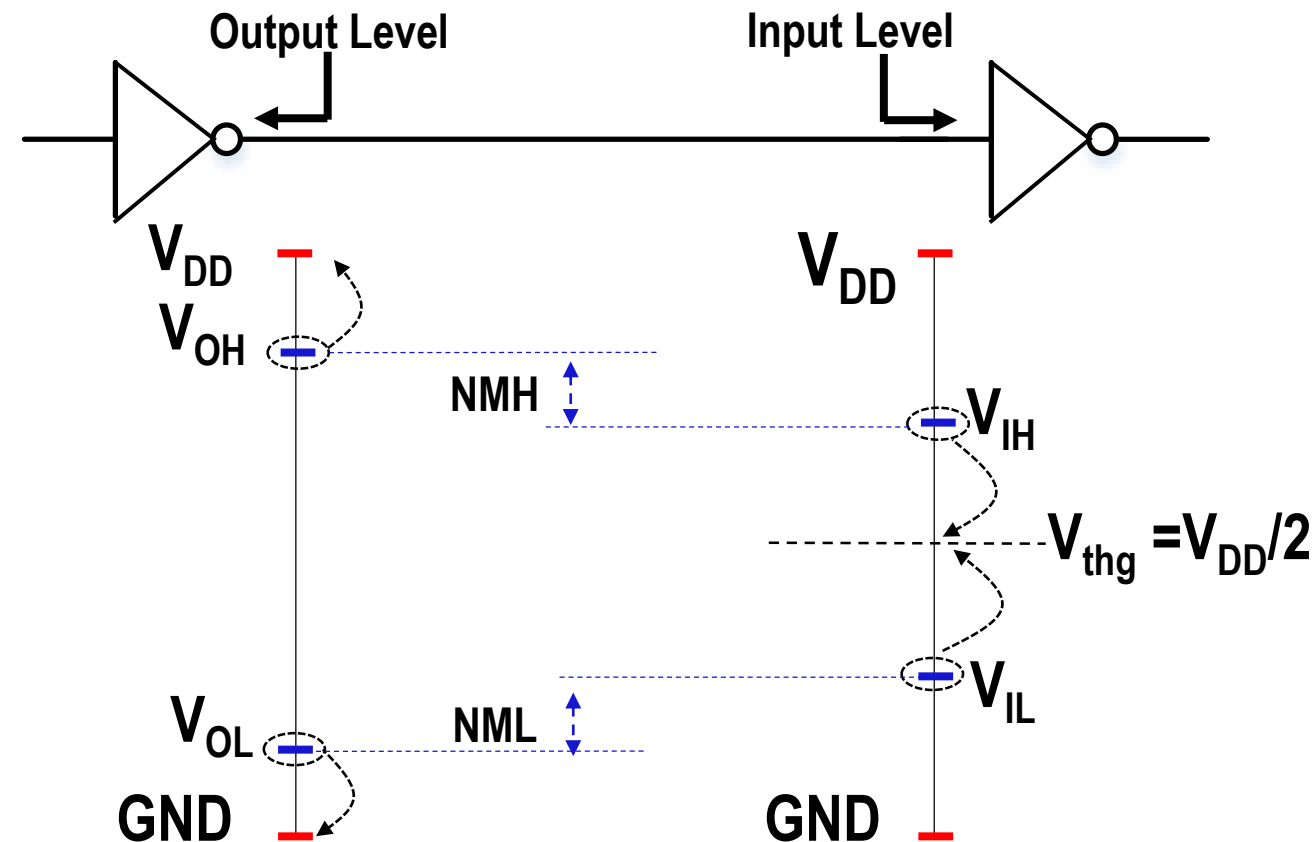
Ideally, $V_{OH} = V_{DD}$, $V_{OL} = GND$

$V_{IH} = V_{IL} = V_{DD}/2$



$NMH = V_{DD}/2$

$NML = V_{DD}/2$



Noise Margin Concept (2/2)



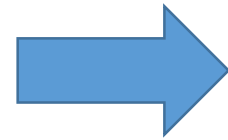
□ Ideal CMOS inverter

□ $NMH = V_{OH} - V_{IH}$

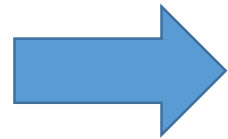
□ $V_{OH} = V_{DD}, V_{IH} = V_{DD}/2$

□ $NML = V_{IH} - V_{OL}$

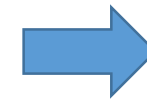
□ $V_{IH} = V_{DD}/2, V_{OL} = GND$



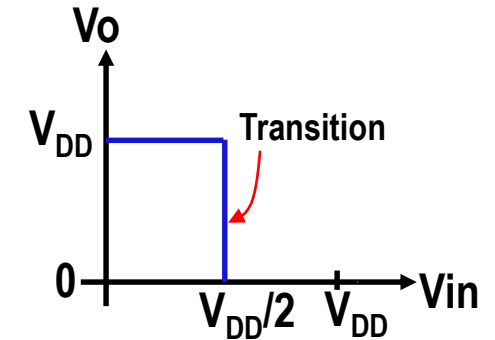
$NMH = V_{DD}/2$



$NML = V_{DD}/2$



$NMH = NML$



□ Practical CMOS inverter

□ To get V_{OH}

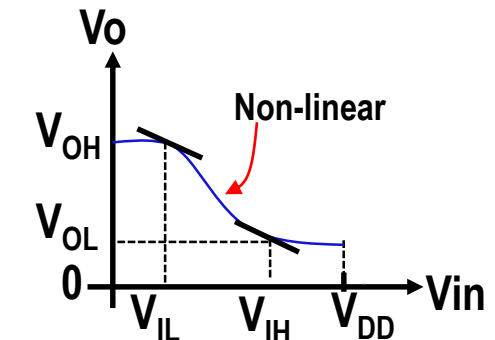
□ Apply $V_{in}=0$, then find $V_o = V_{OH}$

□ To get V_{OL}

□ Apply $V_{in}=V_{DD}$, then find $V_o = V_{OL}$

□ To get V_{IH} and V_{IL}

□ Find relation between V_o and V_{in} , then substitute $\frac{dV_o}{dV_{in}} = -1$.



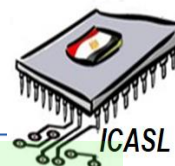
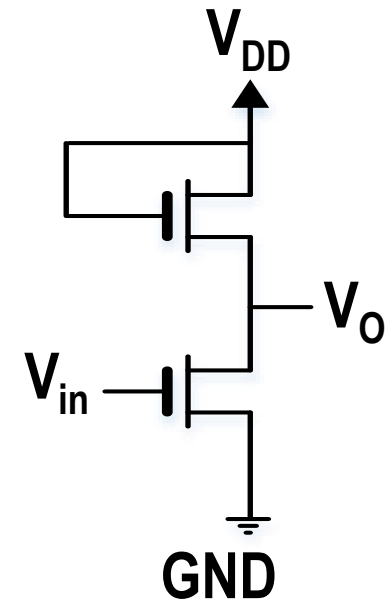
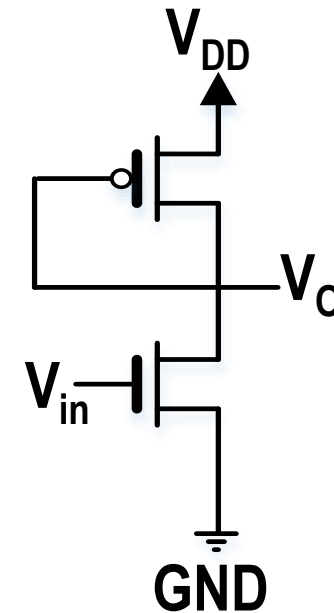
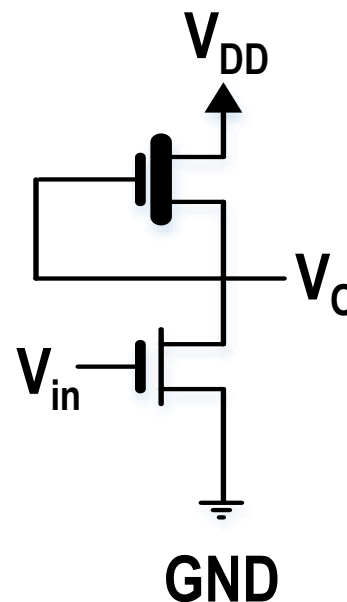
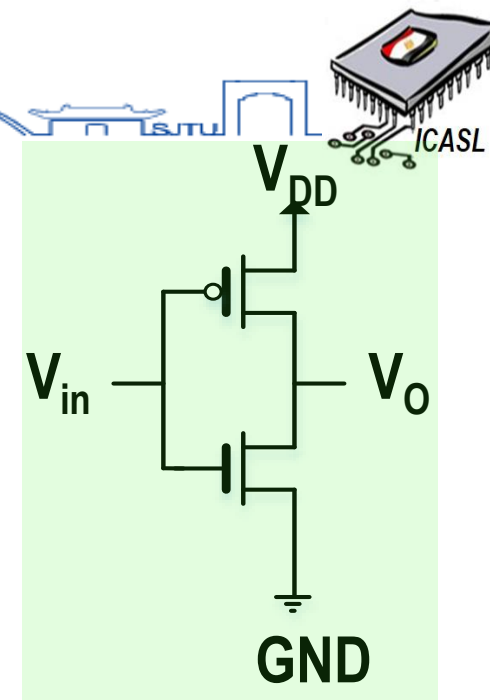
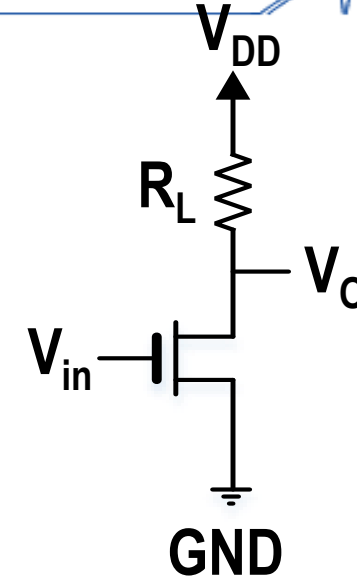
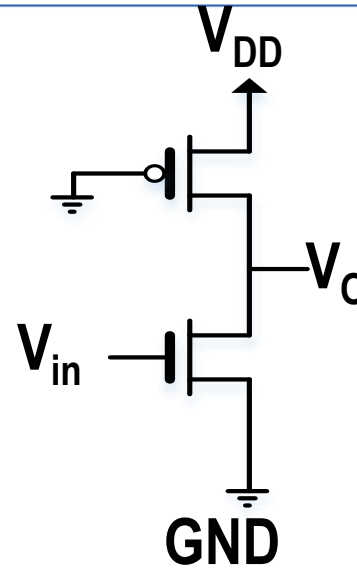
Outlines



- ☐ DC transfer Ch/CS Curve
- ☐ Noise Margin Concept
- ☐ **MOS Inverter Topologies**
- ☐ DC Transfer Characteristics
- ☐ Beta Ratio Effects

MOS Inverter Topologies

- ☒ **CMOS Inverter**
- ☐ **Static Load Inverter**
- ☐ **Pseudo (N-MOS) Inverter**
- ☐ **Saturation (N-MOS) Inverter**
- ☐ **More Saturation (N-MOS) Inverter**
- ☐ **Depletion (N-MOS) Inverter.**



Outlines

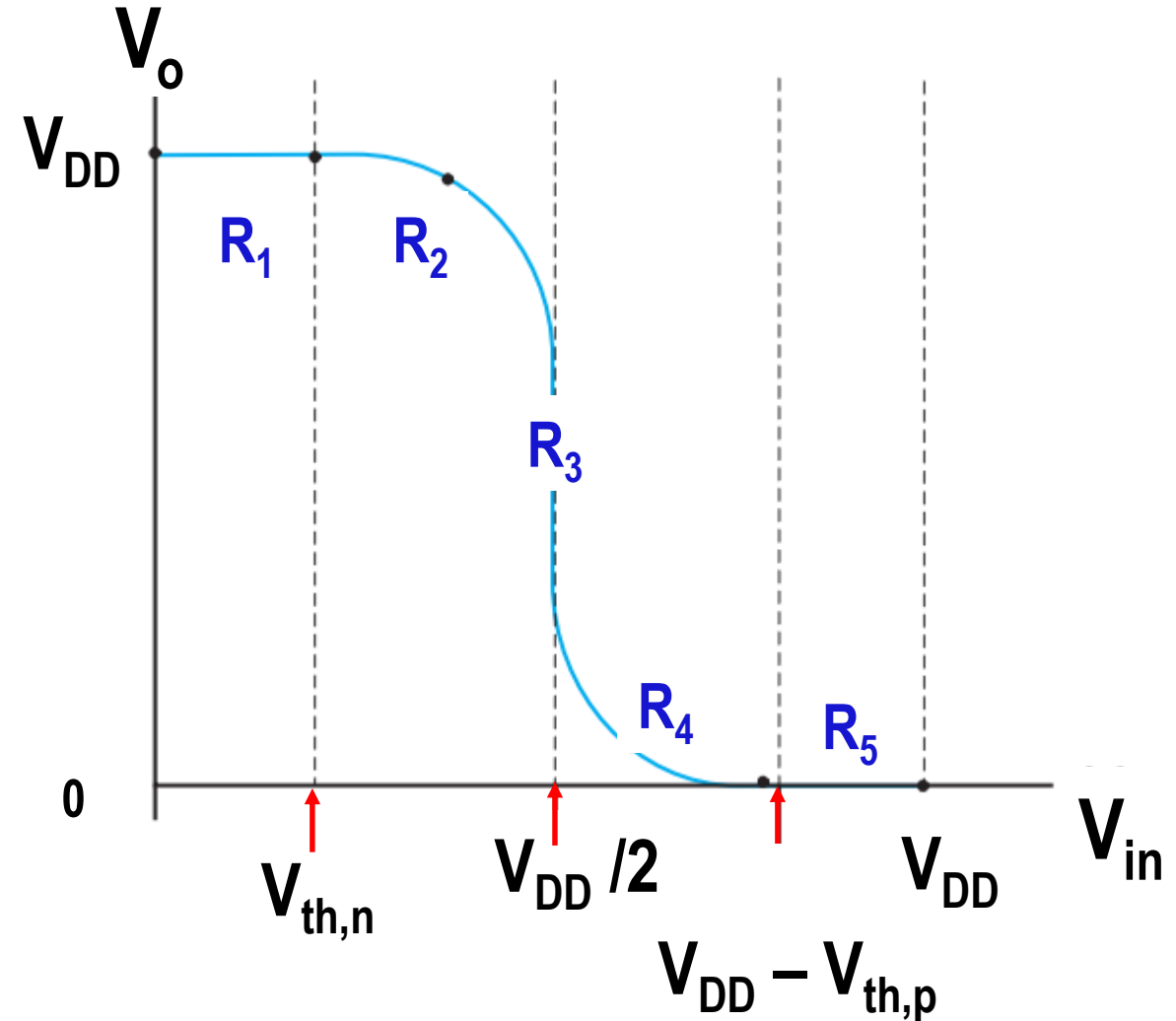
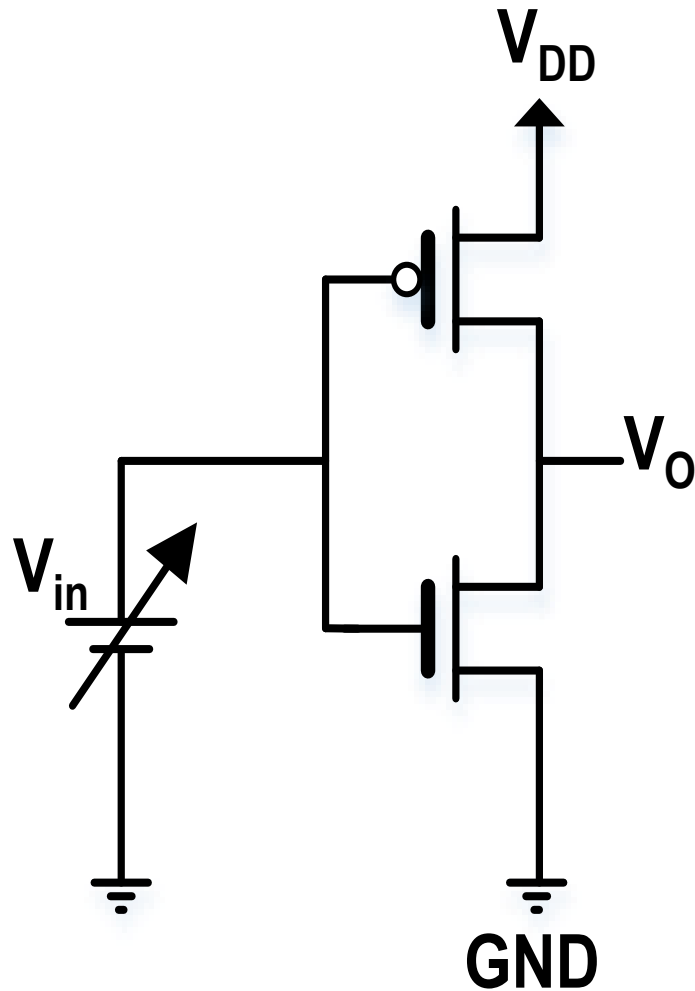


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DC Transfer Characteristics (1/8)

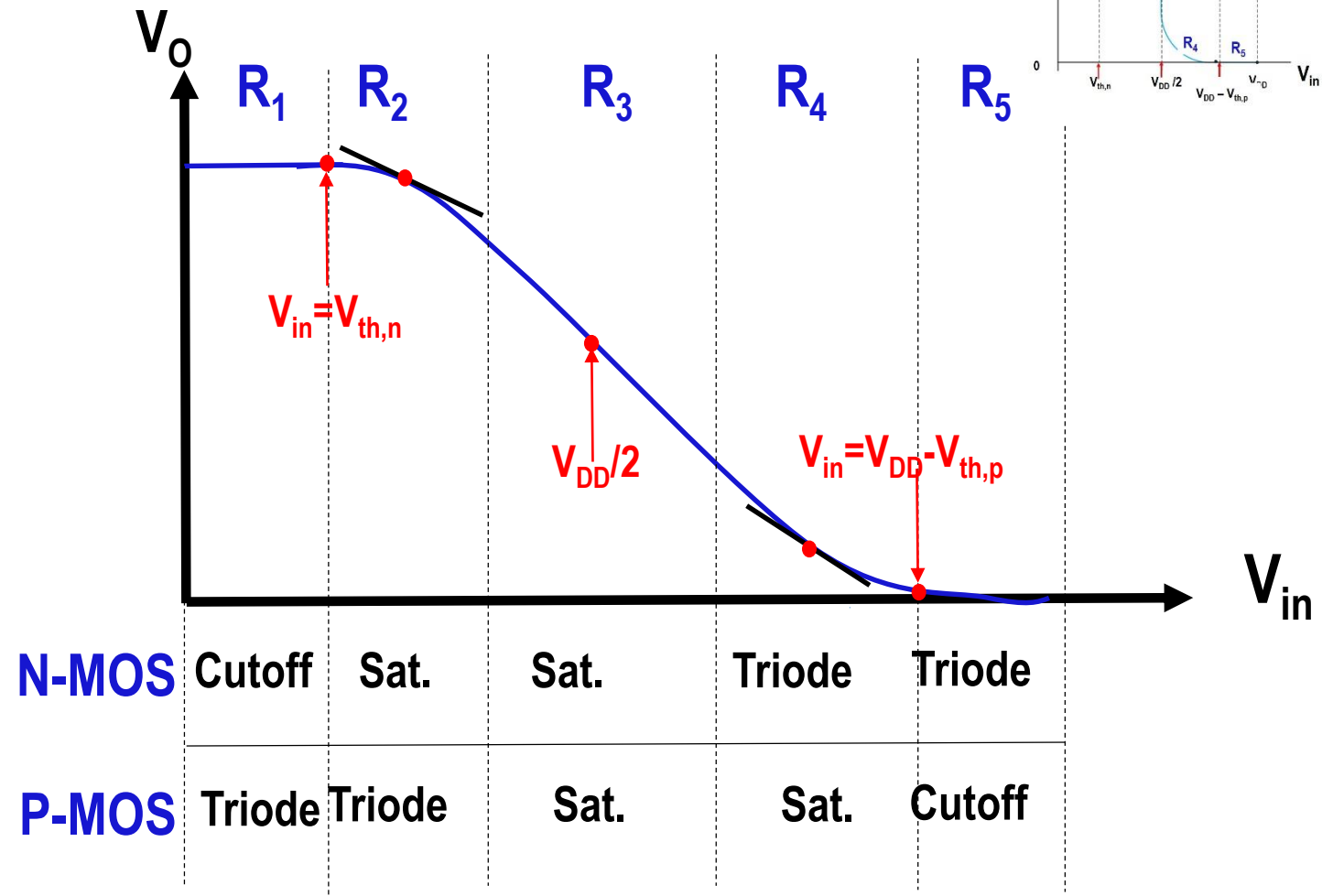
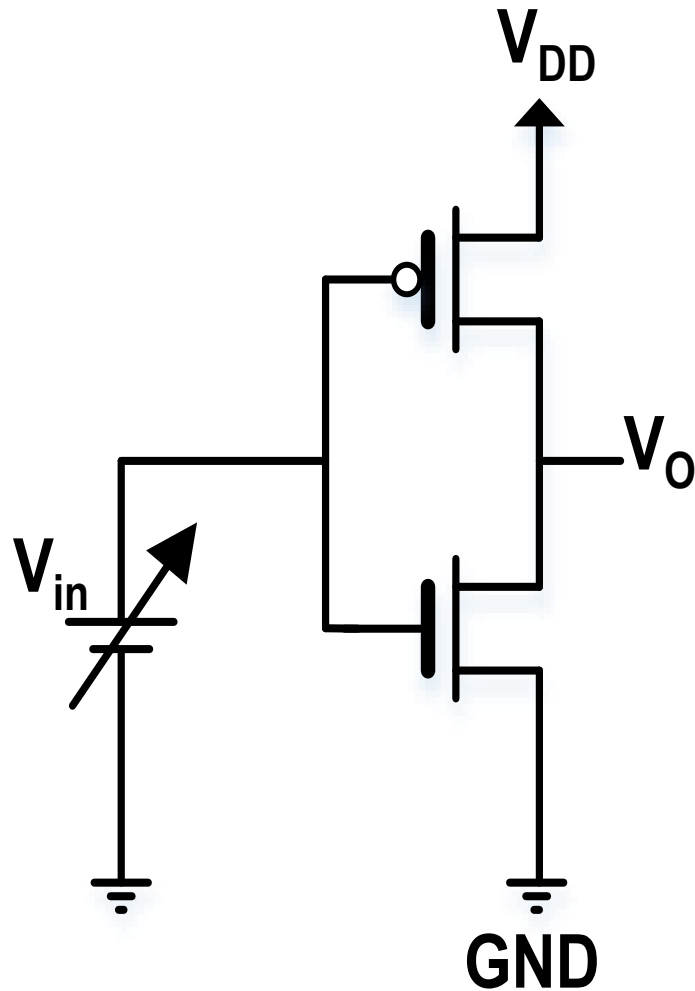


Static CMOS Inverter DC Characteristics



DC Transfer Characteristics (2/8)

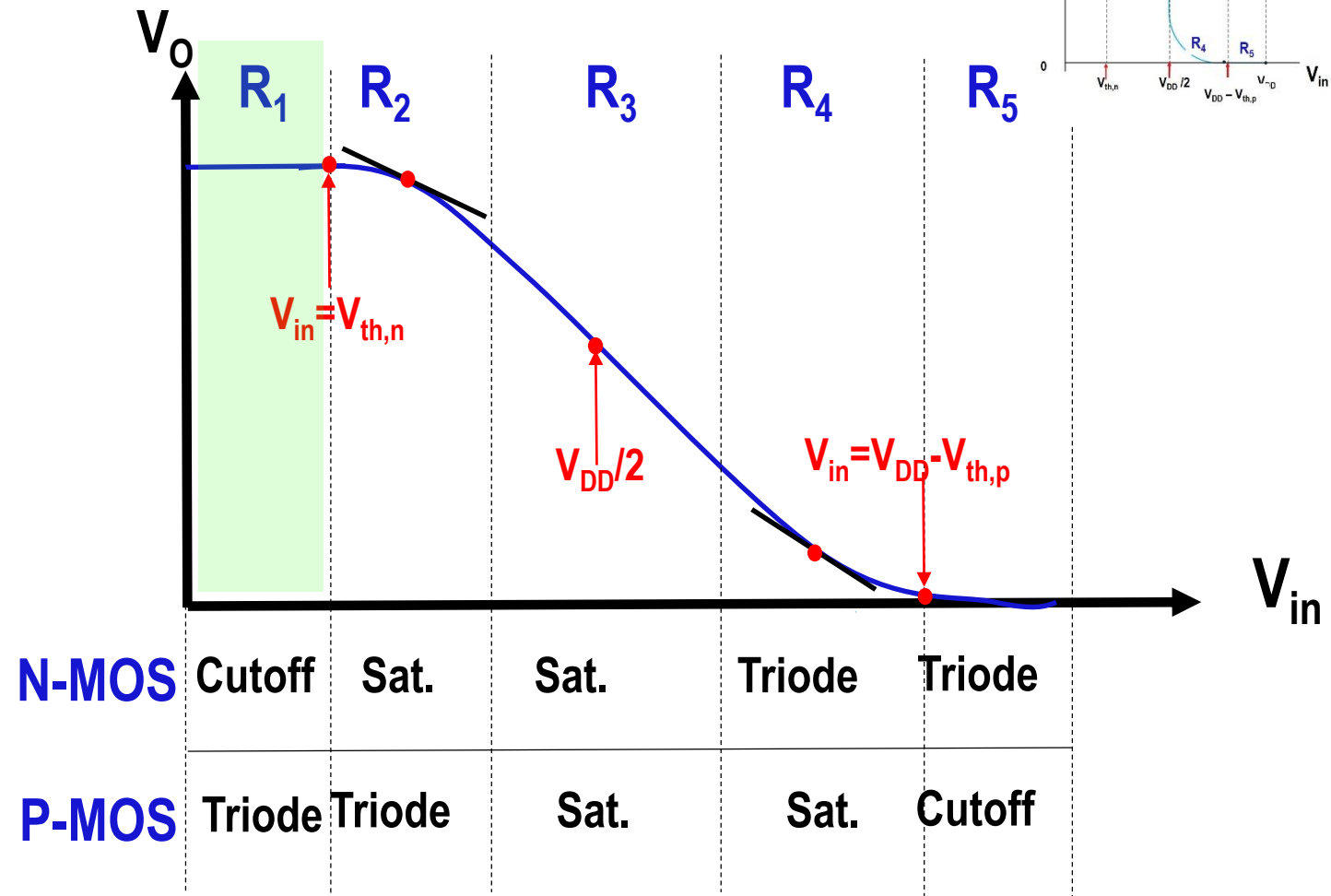
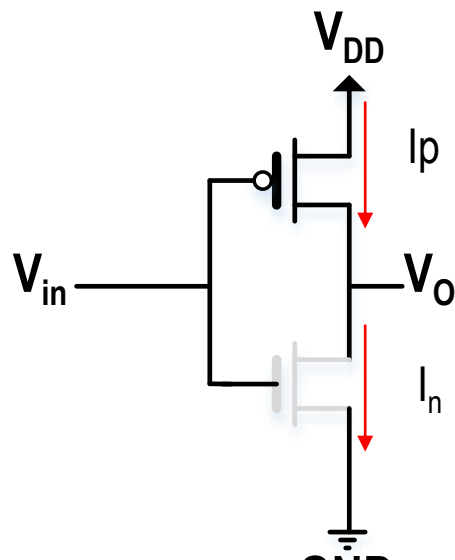
Static CMOS Inverter DC Characteristics



DC Transfer Characteristics (3/8)

Static CMOS Inverter DC Characteristics

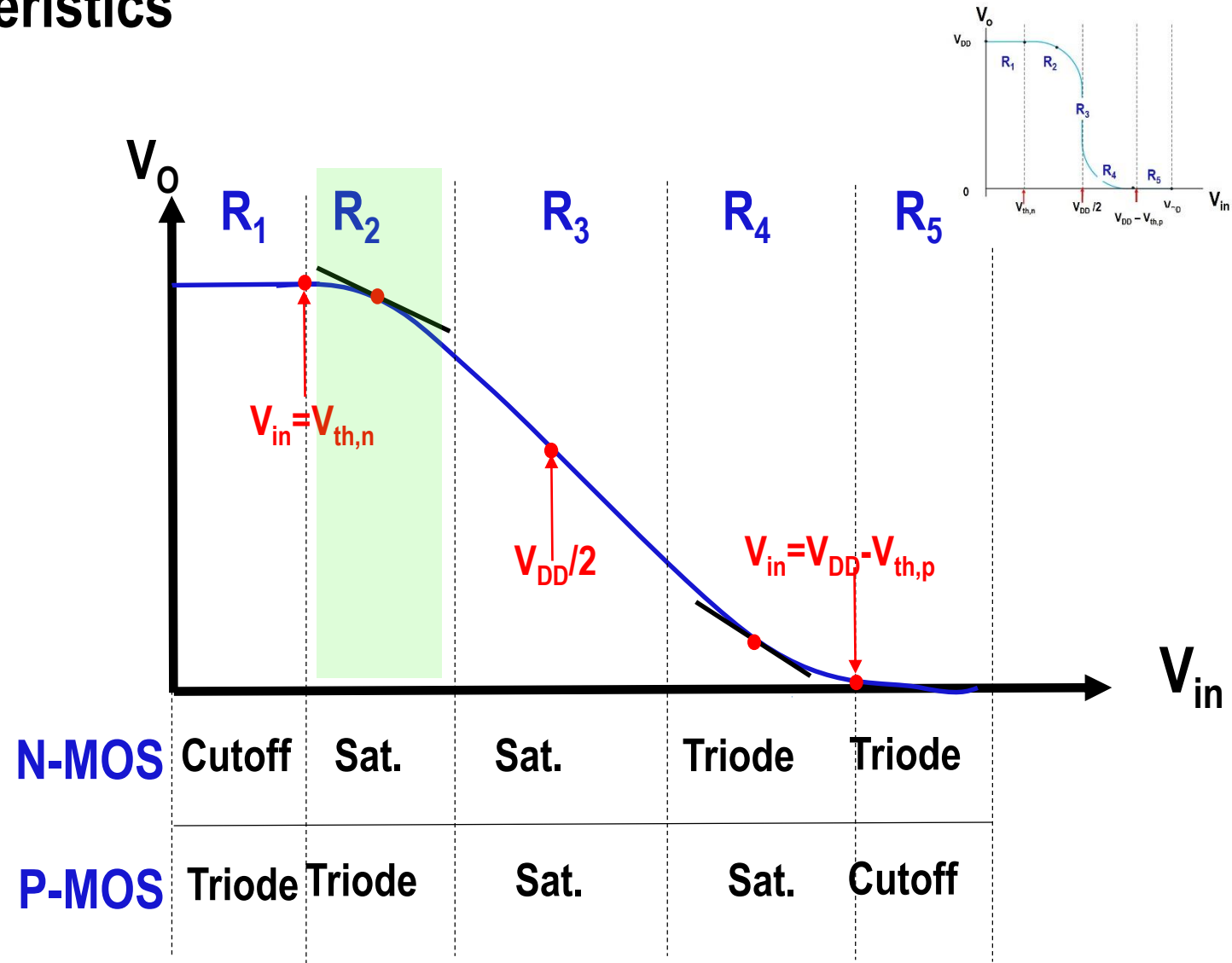
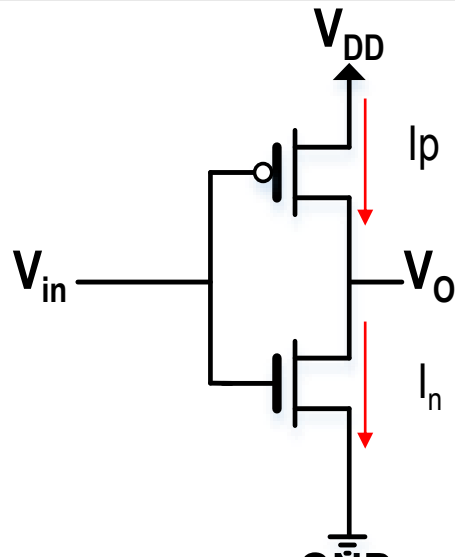
Region (1)	
$V_{in} < V_{th,n}$	
N-MOS	P-MOS
Cutoff (Open Circuit)	Triode
$V_O = V_{DD}$	
$I_{n(cutoff)} = 0 = I_{p(triode)}$	



DC Transfer Characteristics (4/8)

Static CMOS Inverter DC Characteristics

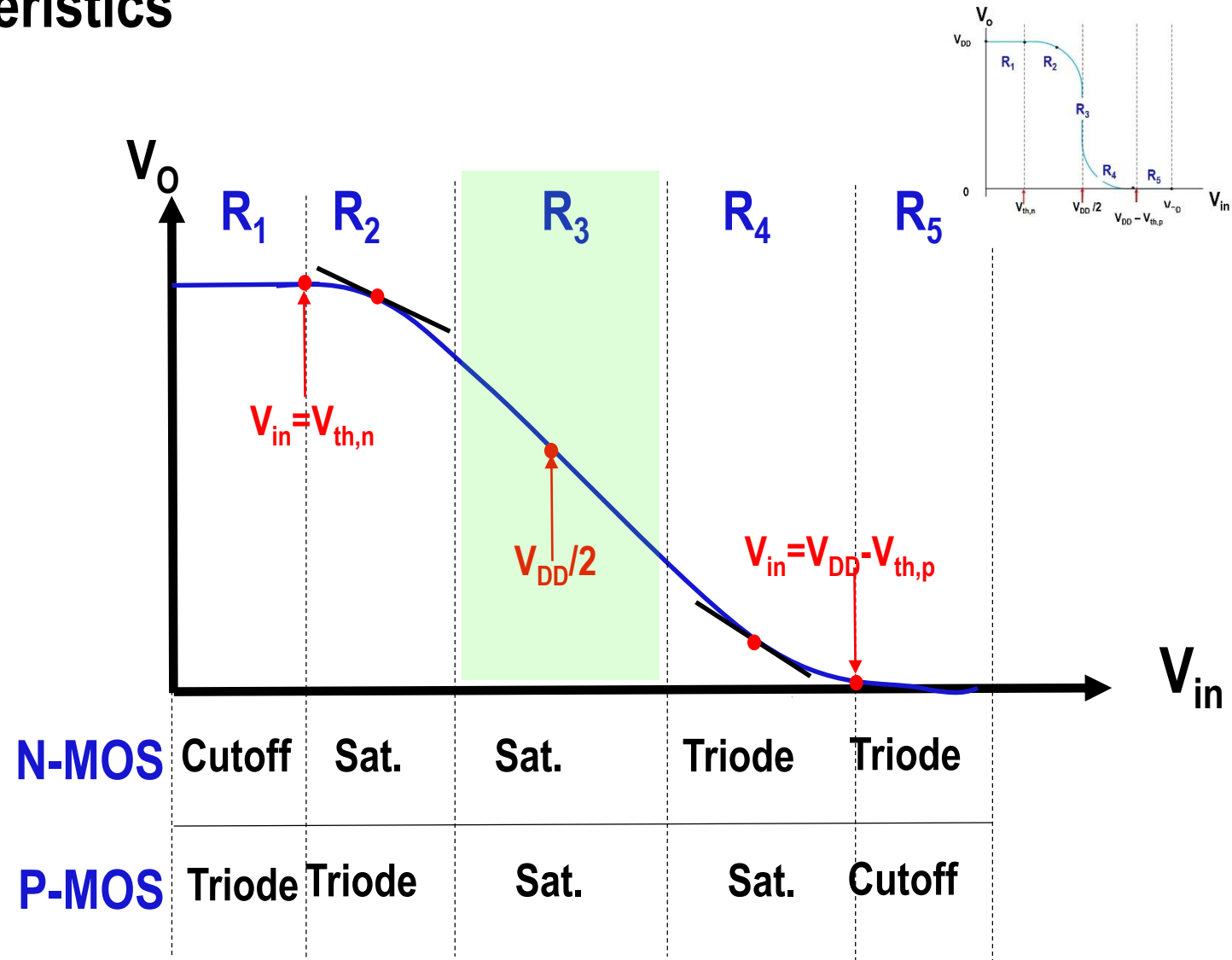
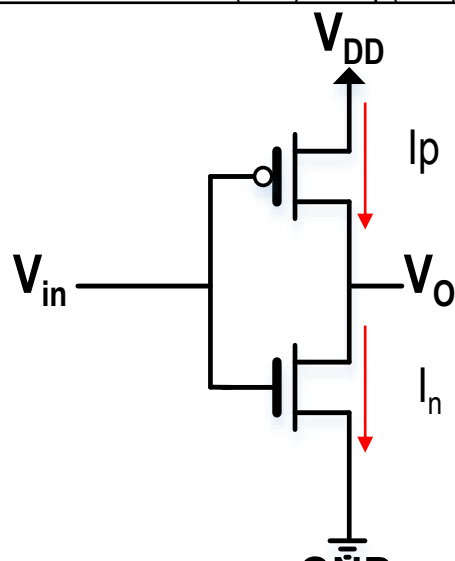
Region (2)	
$V_{DD}/2 > V_{in} \geq V_{th,n}$	
N-MOS	P-MOS
Saturation	Triode
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $V_O \geq V_{in} - V_{th,n}$	$V_{ds,p} > V_{gs,p} - V_{th,p}$ $V_O > V_{in} - V_{th,p}$
$I_{n(Sat.)} = I_{p(triode)}$	



DC Transfer Characteristics (5/8)

Static CMOS Inverter DC Characteristics

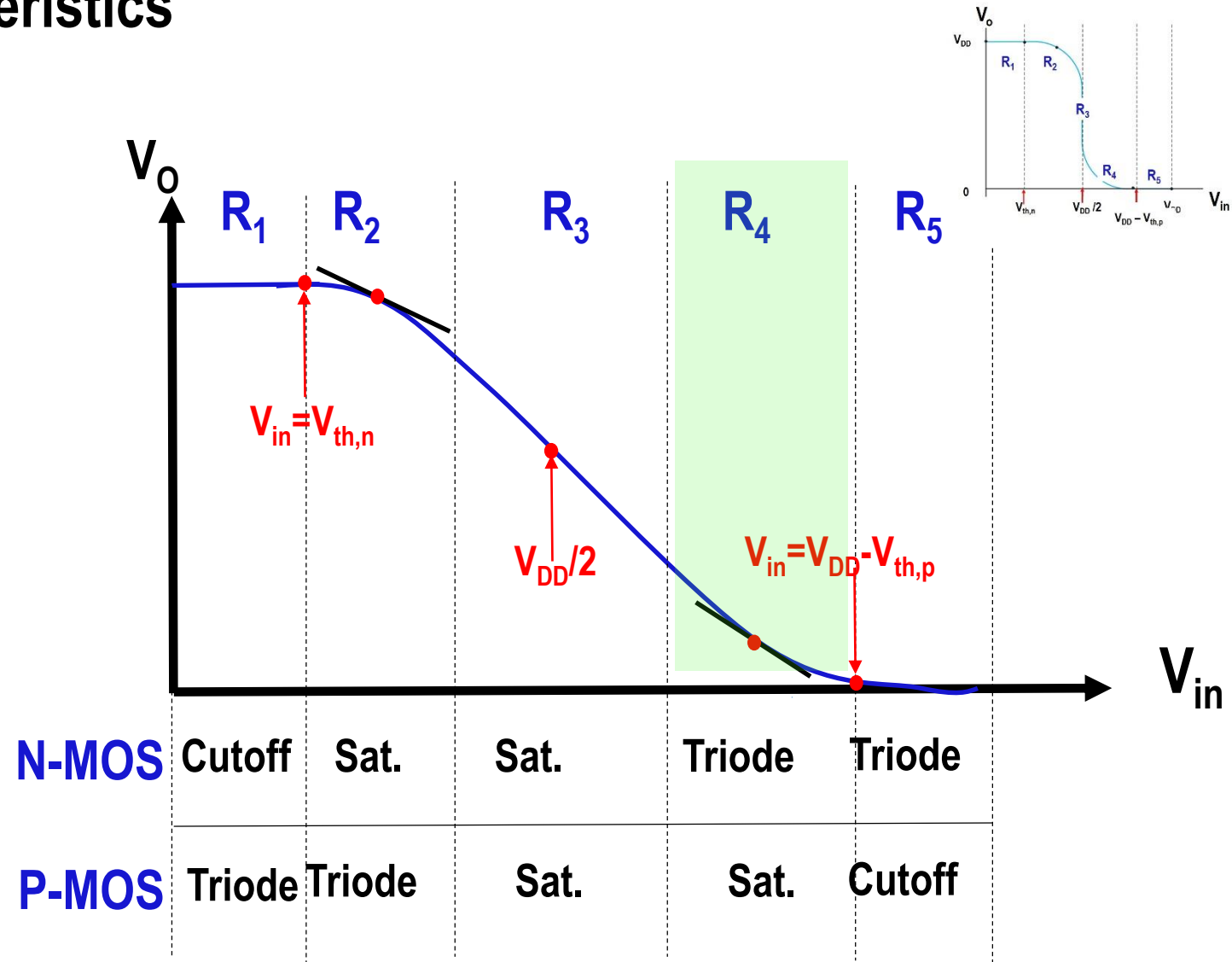
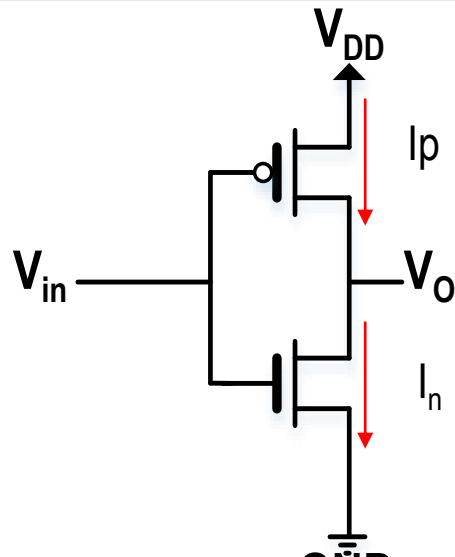
Region (3)	
$V_{in} = V_{DD}/2$	
N-MOS	P-MOS
Saturation	Saturation
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $V_O \geq V_{in} - V_{th,n}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $V_O \leq V_{in} - V_{th,p}$
$I_{n(Sat.)} = I_{p(Sat.)}$	



DC Transfer Characteristics (6/8)

Static CMOS Inverter DC Characteristics

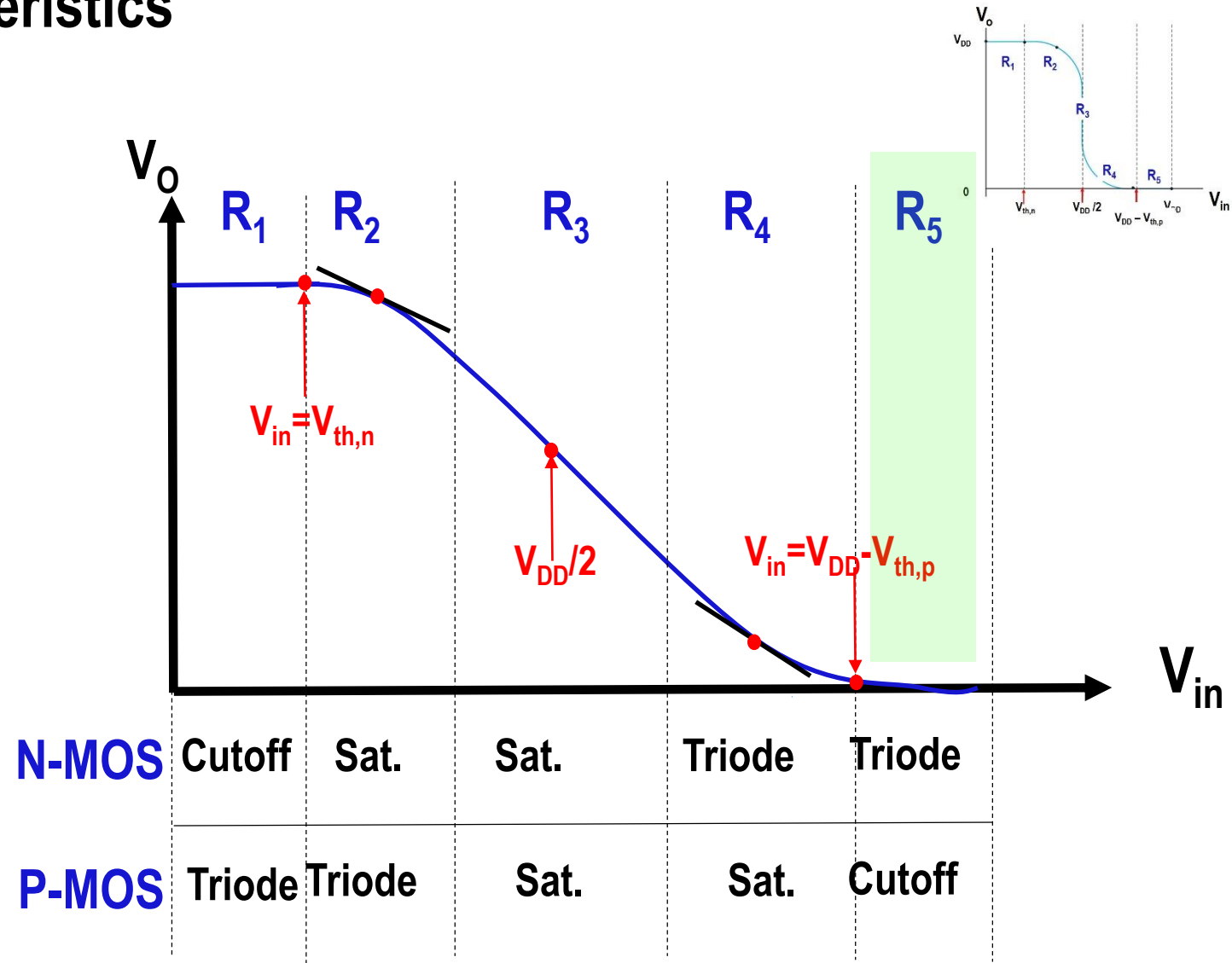
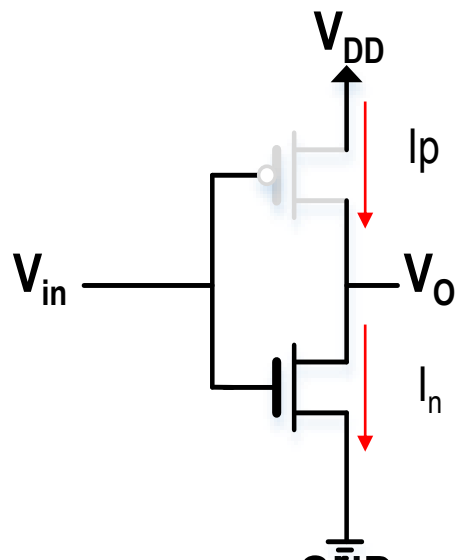
Region (4)	
$V_{DD} - V_{th,p} > V_{in} \geq V_{DD}/2$	
N-MOS	P-MOS
Triode	Saturation
$V_{ds,n} < V_{gs,n} - V_{th,n}$ $V_O < V_{in} - V_{th,n}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $V_O \leq V_{in} - V_{th,p}$
$I_{n(Triode)} = I_{p(Sat.)}$	



DC Transfer Characteristics (7/8)

Static CMOS Inverter DC Characteristics

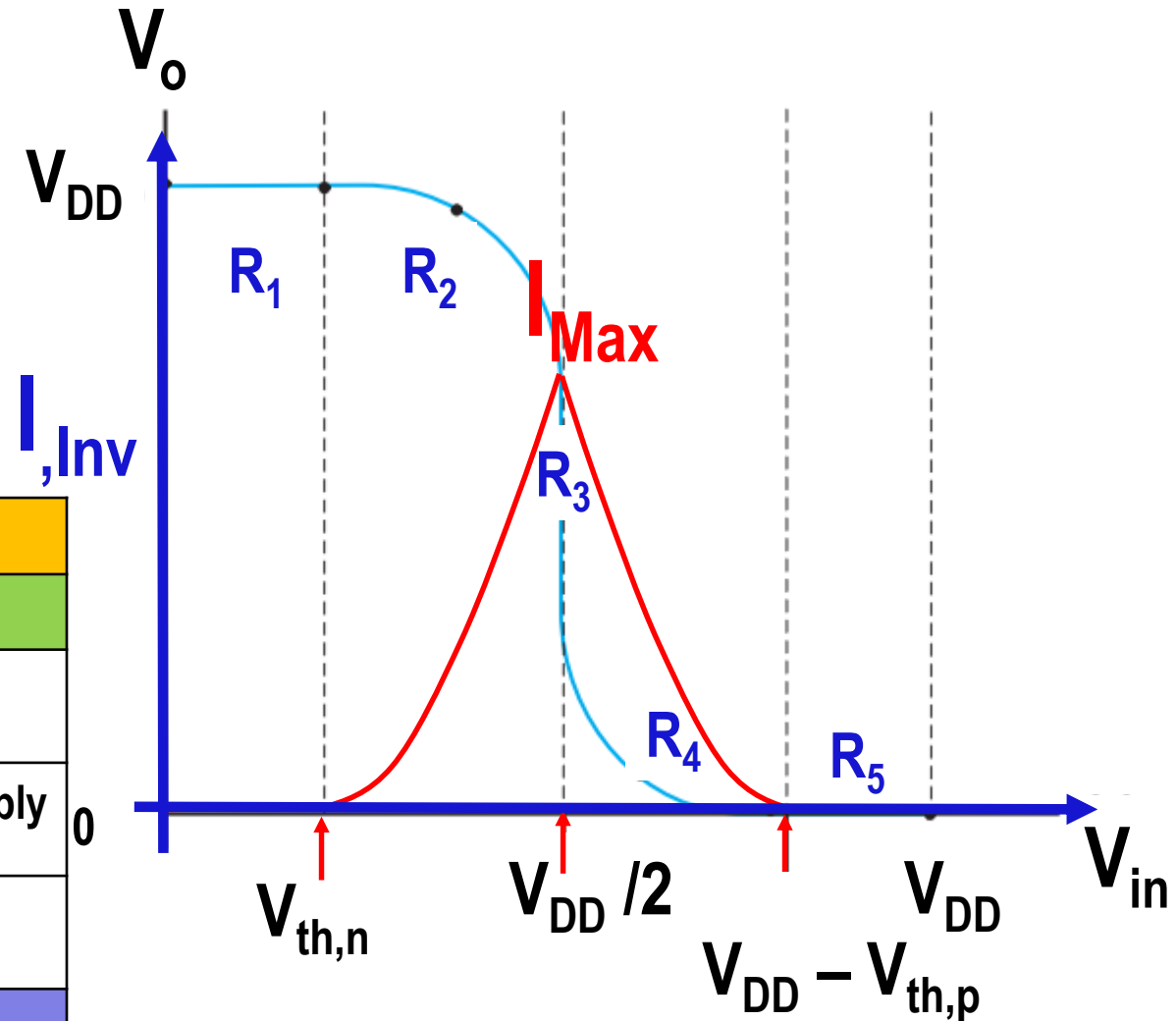
Region (5)	
$V_{in} \geq V_{DD} - V_{th,p}$	
N-MOS	P-MOS
Triode	Cutoff
$V_o = 0$	
$I_{p(Cutoff)} = 0 = I_{n(Triode)}$	



DC Transfer Characteristics (8/8)

- Maximum current will be occurred When both N-MOS and P-MOS are operating in saturation region (region 3)

Region	Condition	N-MOS	P-MOS	Output
1	$0 < V_{in} < V_{th,n}$	Cutoff	Triode	$V_o = V_{DD}$
2	$V_{th,n} \leq V_{in} < V_{DD}/2$	Sat.	Triode	$V_o > V_{DD}/2$
3	$V_{in} = V_{DD}/2$	Sat.	Sat.	V_o drops sharply
4	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{th,p}$	Triode	Sat.	$V_o < V_{DD}/2$
5	$V_{in} \geq V_{DD} - V_{th,p}$	Triode	Cutoff	$V_o = 0$



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Beta Ratio Effects (1/3)



$$B_n = \mu_n C_{ox} \frac{W_n}{L_n}$$

B_n : Gain process factor of n-MOSFET

μ_n : Mobility of Electron

C_{ox} : Gate Oxide Capacitor/unit area

W_n : width of transistor

L_n : length of transistor

$$B_p = \mu_p C_{ox} \frac{W_p}{L_p}$$

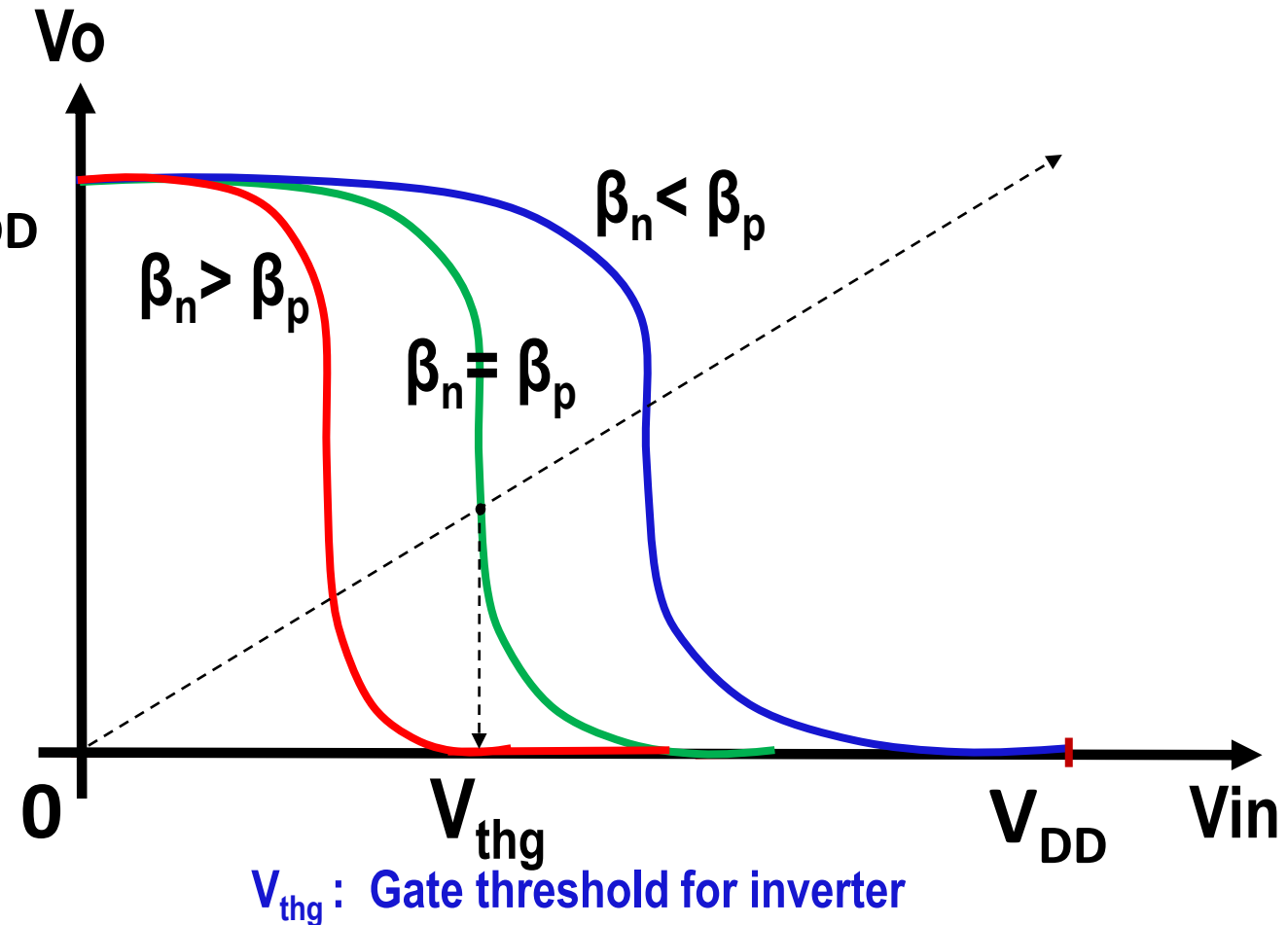
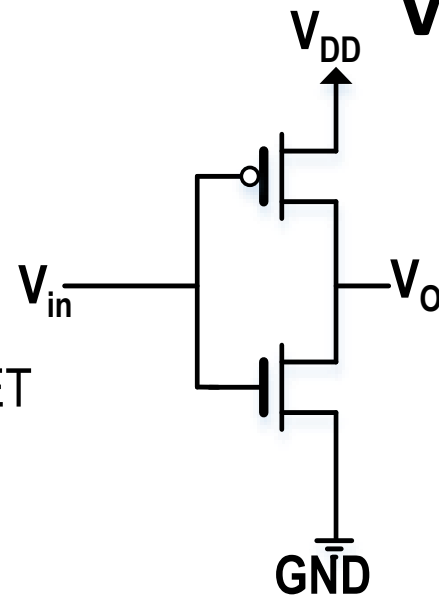
B_p : Gain process factor of p-MOSFET

μ_p : Mobility of Hole

C_{ox} : Gate Oxide Capacitor/unit area

W_p : width of transistor of p-MOSFET

L_p : length of transistor of p-MOSFET



$$V_{thg} = \frac{V_{th,n} + r(V_{DD} + V_{th,p})}{1 + r}$$

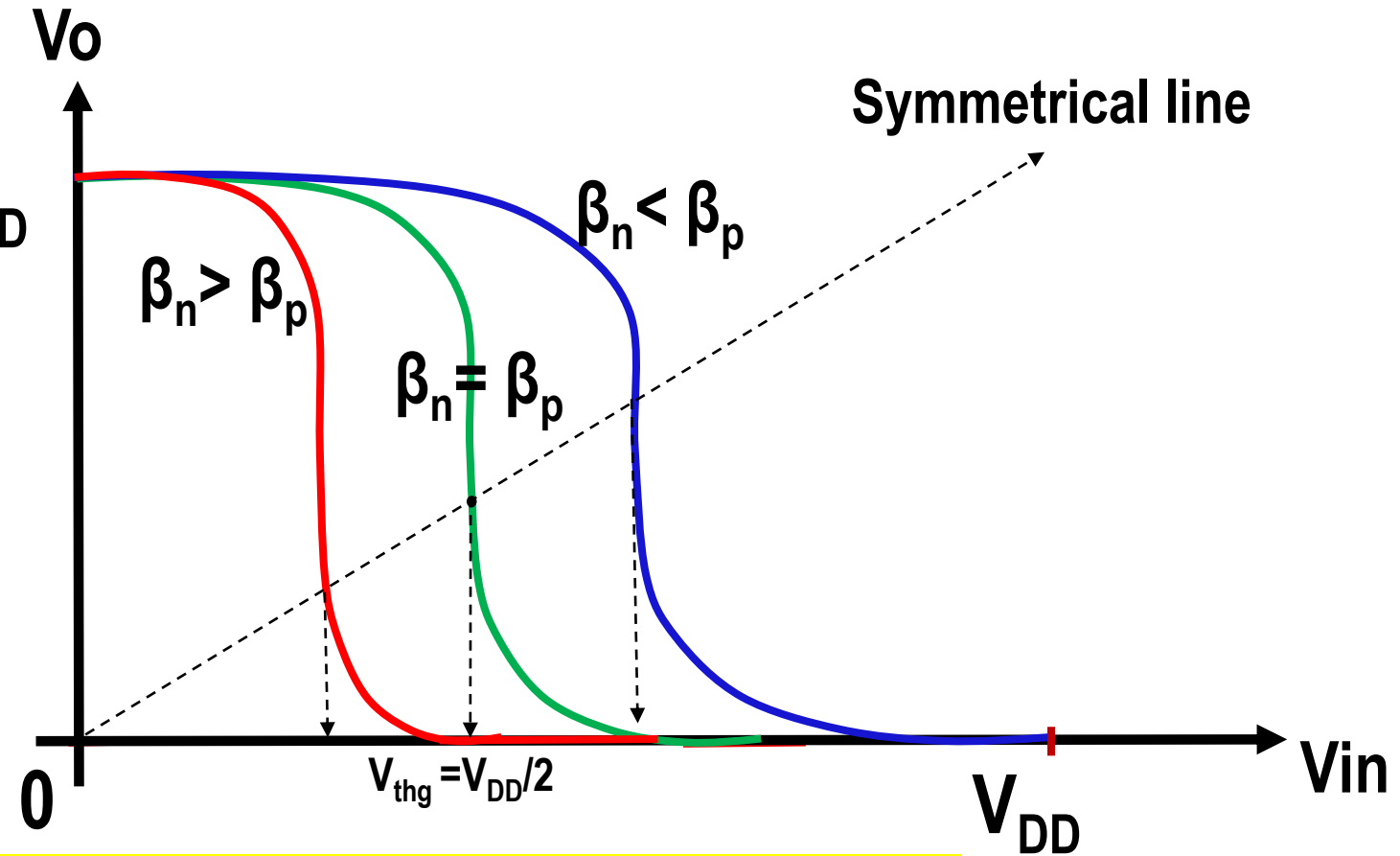
$$\text{Where } r = \sqrt{\frac{\beta_p}{\beta_n}}$$

Beta Ratio Effects (2/3)



Case	Condition	V_{thg}
1	$\beta_n = \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} = V_{DD}/2$
2	$\beta_n > \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} < V_{DD}/2$
3	$\beta_n < \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} > V_{DD}/2$

Inverters with different beta are called *skewed inverters*



To obtain $\beta_n = \beta_p$, $W_p = 2:3 W_n$ & $L_p = L_n$ because $\mu_n = 2:3 \mu_p$

To obtain $\beta_n = \beta_p$, $W_p = 2.5 W_n$ & $L_p = L_n$ because $\mu_n = 2.5 \mu_p$

Symmetrical
Condition

Beta Ratio Effects (3/3)

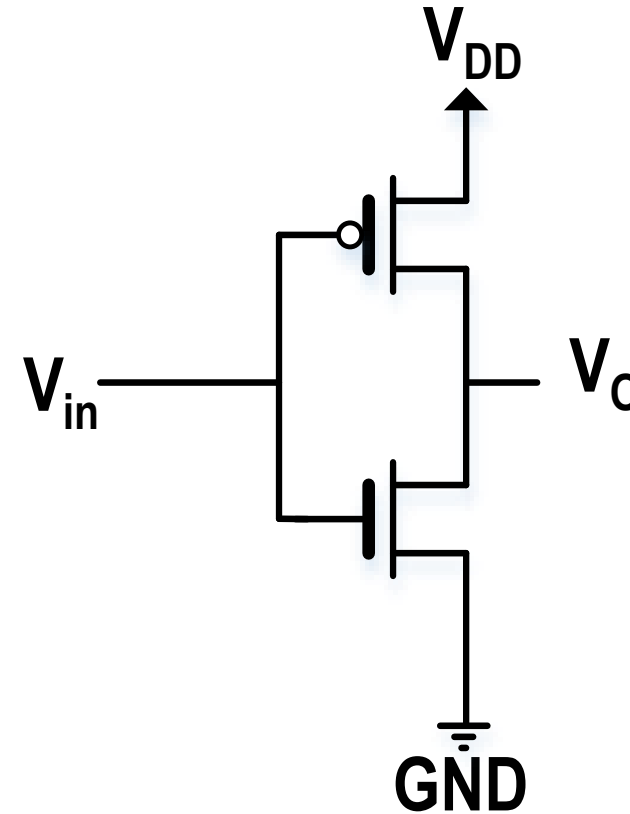


□ Symmetrical condition

□ $\beta_n = \beta_p$, $W_p = 2.5 W_n$ & $L_p = L_n$

□ Minimum Silicon Area

□ $\beta_n > \beta_p$, $W_p = W_n$ & $L_p = L_n$



Thanks!